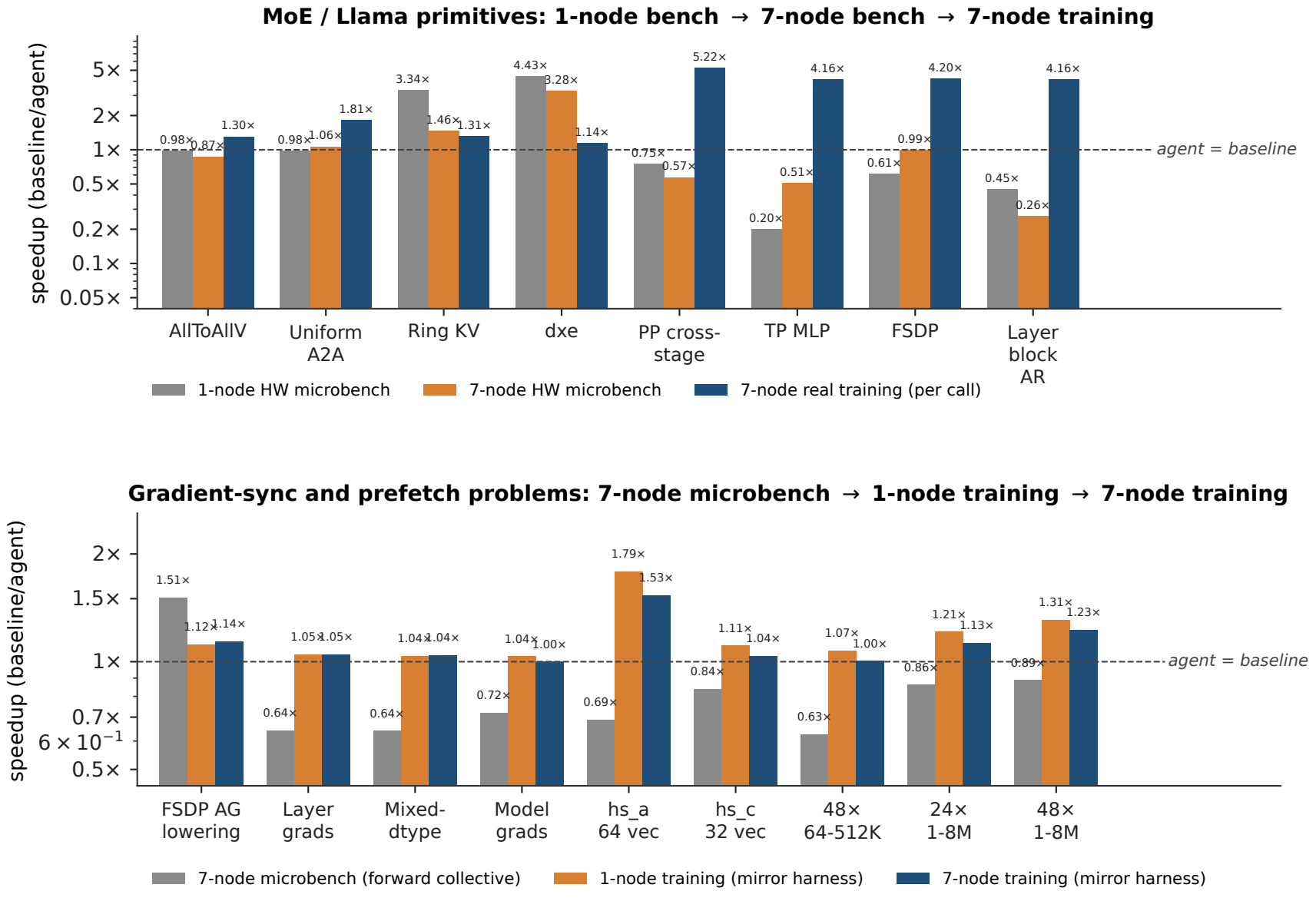
MoE / Llama primitives: 1-node bench → 7-node bench → 7-node training
speedup (baseline/agent)
5×
2×
1×
0.5×
0.2×
0.1×
0.05×
agent = baseline
0.98×
0.87×
1.30×
0.98×
1.06×
1.81×
3.34×
1.46×
1.31×
4.43×
3.28×
1.14×
0.75×
0.57×
5.22×
0.20×
0.51×
4.16×
0.61×
0.99×
4.20×
0.45×
0.26×
4.16×
AllToAllV
Uniform A2A
Ring KV
dxe
PP cross-stage
TP MLP
FSDP
Layer block AR
1-node HW microbench
7-node HW microbench
7-node real training (per call)
Gradient-sync and prefetch problems: 7-node microbench → 1-node training → 7-node training
speedup (baseline/agent)
2×
1.5×
1×
0.7×
6 × 10⁻¹
0.5×
agent = baseline
1.51×
1.12×
1.14×
0.64×
1.05×
1.05×
0.64×
1.04×
1.04×
0.72×
1.04×
1.00×
0.69×
1.79×
1.53×
0.84×
1.11×
1.04×
0.63×
1.07×
1.00×
0.86×
1.21×
1.13×
0.89×
1.31×
1.23×
FSDP AG lowering
Layer grads
Mixed-dtype
Model grads
hs_a 64 vec
hs_c 32 vec
48× 64-512K
24× 1-8M
48× 1-8M
7-node microbench (forward collective)
1-node training (mirror harness)
7-node training (mirror harness)